

# TING-WEI LI

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I am interested in **data-centric machine learning**. I work on data-centric ML and the full data pipeline for modern AI systems, currently focusing on *data selection*, *data attribution*, and *data generation* for LLMs and agentic systems. I have authored 15+ papers at top venues including ICML/NeurIPS/ICLR, with 250+ citations.

## EDUCATION

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- **University of Illinois Urbana-Champaign** Champaign, IL  
**Ph.D. in Computer Science** 08/2024 - Present  
Advisor: Prof. [Hanghang Tong](#)  
GPA: 4.0/4.0; Expected graduation: 12/2028
- **University of Michigan, Ann Arbor** Ann Arbor, MI  
**M.Sc. in Electrical and Computer Engineering** 08/2021 - 04/2023  
GPA: 4.0/4.0
- **National Taiwan University** Taipei, Taiwan  
**B.S.Eng in Electrical Engineering** 09/2016 - 06/2020  
GPA: 3.9/4.3

## EXPERIENCE

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- **Meta Platforms, Inc.**  
**Part-Time Student Researcher** 08/2026 - Present, Remote, IL
  - Developed efficient data selection methods tailored for large-scale Ads recommendation models.
  - Mentors: Amit Anand Amlesahwaram, Bill Huang**Research Intern** 05/2026 - 08/2026, Sunnyvale, CA
  - Developed gradient-based data attribution methods tailored for large-scale Ads recommendation models.
  - Mentors: Amit Anand Amlesahwaram, Bill Huang
- **Amazon.com, Inc.**  
**Applied Scientist Intern** 05/2025 - 08/2025, Seattle, WA
  - Designed an efficient multi-task learning framework for a large-scale video platform.
  - Mentors: Cody Severinski, Changhe Yuan
- **AT&T Inc.**  
**Chief Data Office Intern (Research Intern)** 09/2023 - 11/2023, Bedminster, NJ
  - Developed interpretable graph-based models for large-scale customer behavior analysis.
  - Mentors: Qiong Wu, Zhengyi Zhou, Eric Zavesky

## SELECTED PUBLICATIONS

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1. **Ting-Wei Li**, Yuanchen Bei, Xiao Lin, Hanghang Tong. “*Beyond LLM-Based Reasoning: Lightweight GNNs for Agent Failure Attribution*”. In **Conference on Neural Information Processing Systems (NeurIPS)** 2026. [\[Paper\]](#).
2. **Ting-Wei Li**, Sirui Chen, Jiaru Zou, Yingbing Huang, Tianxin Wei, Jingrui He, Hanghang Tong. “*EvoSelect: Data-Efficient LLM Evolution for Targeted Task Adaptation*”. In **Conference on Language Modeling (COLM)** 2026. [\[Paper\]](#).
3. Tianxin Wei\*, **Ting-Wei Li\***, Zhining Liu\* et al. “*A Survey of Agentic Reasoning for Large Language Models: Towards Recursively Self-Improving and Collective Agents*”. In **Transactions on Machine Learning Research (TMLR)** 2026 (Hugging Face Weekly #1 Paper). [\[Paper\]](#).

4. Xuying Ning, Dongqi Fu, Tianxin Wei, Mengting Ai, Jiaru Zou, **Ting-Wei Li**, Hanghang Tong, Yada Zhu, Hendrik Hamann, Jingrui He. “*MC-Search: Evaluating and Enhancing Multimodal Agentic Search with Structured Long Reasoning Chains*”. In **International Conference on Learning Representations (ICLR)** Oral 2026. [Paper].
5. Ruizhong Qiu\*, **Ting-Wei Li\***, Gaotang Li, Hanghang Tong. “*Graph Homophily Booster: Rethinking the Role of Discrete Features on Heterophilic Graphs*”. In **International Conference on Learning Representations (ICLR)** 2026. [Paper].
6. Hyunsik Yoo, **Ting-Wei Li**, SeongKu Kang, Zhining Liu, Charlie Xu, Qilin Qi, Hanghang Tong. “*Continual Low-Rank Adapters for LLM-based Generative Recommender Systems*”. In **International Conference on Learning Representations (ICLR)** 2026. [Paper].
7. **Ting-Wei Li**, Ruizhong Qiu, Hanghang Tong. “*Graph Data Selection for Domain Adaptation: A Model-Free Approach*”. In **Conference on Neural Information Processing Systems (NeurIPS)** 2025. [Paper] [Poster].
8. Junwei Deng\*, **Ting-Wei Li\***, Shiyuan Zhang, Shixuan Liu, Yijun Pan, Hao Huang, Xinhe Wang, Pingbang Hu, Xingjian Zhang, Jiaqi W Ma. “*dattri: A Library for Efficient Data Attribution*”. In **Conference on Neural Information Processing Systems (NeurIPS) Spotlight** 2024. [Paper] [Code] [Poster].
9. Junwei Deng\*, **Ting-Wei Li\***, Shichang Zhang, Jiaqi Ma. “*Efficient Ensembles Improve Training Data Attribution*”. In **Transactions on Machine Learning Research (TMLR)** 2024. [Paper].
10. **Ting-Wei Li**, Qiaozhu Mei, Jiaqi Ma. “*A Metadata-Driven Approach to Understand Graph Neural Networks*”. In **Conference on Neural Information Processing Systems (NeurIPS)** 2023. [Paper] [Code] [Poster].

(\* indicates equal contribution)

## PREPRINTS AND SUBMISSIONS

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1. **Ting-Wei Li**, Ernie Chang, Yuanchen Bei, Narine Kokhlikyan, Hanghang Tong. “*HyperBridge: Effectively Adapting Hypernetworks across LLM Architectures*”. In **Arxiv (In submission to ICLR)** 2026.
2. Xuying Ning, Katherine Tieu, Dongqi Fu, Tianxin Wei, Zihao Li, Yuanchen Bei, Jiaru Zou, Mengting Ai, Zhining Liu, **Ting-Wei Li** et al. “*Code as Agent Harness: Toward Executable, Verifiable, and Stateful Agent Systems*”. In **Arxiv** 2026. [Paper].
3. Xinrui He, **Ting-Wei Li**, Junting Wang, Mengting Ai, Xinyu He, Hanghang Tong, Jingrui He. “*Harnessing Image Question Dependence for Better VLM Test-time Reinforcement Learning*”. In **Arxiv (In submission to ICLR)** 2026. [Paper].
4. Ruizhong Qiu\*, **Ting-Wei Li\***, Gaotang Li\*, Tianxin Wei, Jingrui He, Hanghang Tong. “*Efficient Inference Scaling for Safety Assurance*”. In **Arxiv** 2025. [Paper].
5. Junwei Deng\*, Yuzheng Hu\*, Pingbang Hu\*, **Ting-Wei Li\***, Shixuan Liu\* et al. “*A Survey of Data Attribution: Methods, Applications, and Evaluation in the Era of Generative AI*”. In **SSRN (In submission to JAIR)** 2025. [Paper].

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## SKILLS

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- **Programming:** Python, C++, Bash
- **ML & LLM Frameworks:** PyTorch, Hugging Face, vLLM, TRL, verl, PEFT
- **AI-Native Development:** Claude Code, Claude CLI, Cursor

## SELECTED AWARDS & HONORS

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- **Scholar Award**, Conference on Neural Information Processing Systems 2023
- **Graduate Student Research Assistant Tuition Waiver**, University of Michigan 2023

INVITED TALKS

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- **Agentic Reasoning for Large Language Models**, NLP Academic Exchange Platform 2026
- **Data Attribution for GenAI Copyright**, Data Attribution Reading Group 2024

PROFESSIONAL SERVICES

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- **Conference Reviewer**
  - ACM Web Conference (WWW) 2027
  - IEEE/CVF International Conference on Computer Vision (ICCV) 2025
  - Conference on Neural Information Processing Systems (NeurIPS) 2025, 2026
  - International Conference on Learning Representations (ICLR) 2025, 2026, 2027
  - International Conference on Machine Learning (ICML) 2024, 2025, 2026
  - IEEE International Conference on Big Data (Big Data) 2023
  - IEEE International Conference on Data Mining Workshop on Machine Learning on Graphs (ICDM MLoG) 2023
- **Program Committee**
  - AAAI Conference on Artificial Intelligence (AAAI) 2025, 2026, 2027
  - IEEE International Conference on Data Science and Advanced Analytics (DSAA) 2024
- **Organizer Team**
  - Data Foundations of AI Seminar Coordinator
  - Data Attribution Reading Group Speaker, Organization Member

MENTORING EXPERIENCE

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- **Mr. Wei-Chi Wu (Exchange B.S. Student, UIUC CS)** 2026 - Now
  - Hypernetworks for non-text data and domain adaptation.
- **Ms. Thota Hima (B.S. Student, UIUC CS)** 2026 - Now
  - Evolving semantic tokenization for LLM-based continual recommendation systems.
- **Ms. Jianing Li (B.S. Student, UIUC CS)** 2025 - 2026
  - Graph-augmented LLM-based recommendation systems.

TEACHING EXPERIENCE

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- University of Illinois Urbana-Champaign, **CS 412: Introduction to Data Mining** Spring 2026